

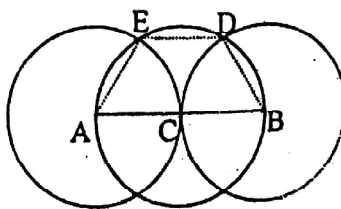
**AMTI (NMTC) - 2004**  
**BHASKARA CONTEST - JUNIOR LEVEL**

1. Given the sequence a, ab, aab, aaabb, aaabbb, .... upto 2004 terms, the total number of times a's and b's are used from 1 to 2004 terms are  
 (A) 2004 a's and 2003 b's (B) 4008 a's and b's  
 (C)  $1002 \times 1003$  a's and  $(1002)^2$  b's (D)  $1003^2$  a's and  $1002 \times 1003$  b's
2. The number of two digit numbers divisible by the product of the digits is  
 (A) 5 (B) 8 (C) 14 (D) 33
3. Given that  $(a - 5)^2 + (b - c)^2 + (c - d)^2 + (b + c + d - 9)^2 = 0$  then  $(a + b + c)(b + c + d)$  is  
 (A) 0 (B) 11 (C) 20 (D) 99
4.  $x^4 - y^4 = 15$ , x and y are positive integers. Then  $x^4 + y^4$  is  
 (A) 17 (B) 31 (C) 32 (D) 113
5. In this addition each letter represents a different digit. Which is the absent digit ?

$$\begin{array}{r} \text{ABCD} \\ + \text{BCD} \\ \hline \text{GHIJK} \end{array}$$

- (A) 1 (B) 3 (C) 4 (D) 5
6. Five children each owned a different number of rupees. The ratio of any one's fortune to the fortune of every child poorer than himself was an integer. The combined fortune of the children was 847 rupees. The least number of rupees that a child had was  
 (A) 12 Rs. (B) 10 Rs. (C) 7 Rs. (D) 5 Rs.
7. A number with 8 digits is multiple of 73 and also a multiple of 137. The second digit from the left equals 7. Then the 6<sup>th</sup> digit from the left equals  
 (A) 1 (B) 7 (C) 9 (D) can be any digit
8. Let n be the least positive integer such that  $1260n$  is the cube of a natural number. Then n satisfies  
 (A)  $1 < n < 50$  (B)  $50 < n < 100$   
 (C)  $100 < n < 1000$  (D)  $1000 < n < 10000$
9. If  $(43)_x$  in base x number system is equal to  $(34)_y$  in base y number system the possible value for  $x + y$  is  
 (A) 16 (B) 14 (C) 12 (D) 10
10. In each of the following 2003 fractions the sum of the numerator and denominator equals 2004 :  $\left[ \frac{1}{2003}, \frac{2}{2002}, \frac{3}{2002}, \dots, \frac{2003}{1} \right]$ . The number of fractions  $< 1$  which are irreducible (no common factor between numerator and denominator) is  
 (A) 664 (B) 332 (C) 1002 (D) 1001
11. For how many integers n is  $\sqrt{9 - (n + 2)^2}$  a real number ?  
 (A) 3 (B) 5 (C) 7 (D) infinitely many
12. During holidays, five people A, B, C, D and E went swimming regularly. Each time they went, exactly one of them was missing. A went the least number of times (5 times) and E most often (8 times). What can we say about the number of times B, C and D went ?  
 (A) each went six times  
 (B) each went seven times  
 (C) 2 went 6 times and one went 7 times  
 (D) 2 went 7 times and 1 went 6 times

13. Let  $A = \{a, b, c\}$  and  $B = \{a, b, d, e, f\}$ . How many sets  $C$  consisting of characters from the English alphabet can be constructed so that  $C \subseteq B$  and such that  $A \cap C$  has one element and  $C \not\subseteq A$ .  
 (A) 16 (B) 14 (C) 8 (D) 6
14. The sum of all angles except one of a convex polygon is  $2190^\circ$ . (where the angles are less than  $180^\circ$ ). Then the possible number of sides of the polygon is  
 (A) 13 (B) 15 (C) 17 (D) 19
15. In a right angled triangle with legs 4 and 8, the area of the largest square that can be inscribed in the triangle is  
 (A)  $\frac{8}{3}$  (B)  $\frac{4}{3}$  (C)  $\frac{16}{9}$  (D)  $\frac{16}{9}$
16. Two circles with centres A and B and radius 2 touch each other externally at C. A third circle with center C and radius 2 meets the other two at D, E (see the figure). Then area ABDE is



- (A)  $3\sqrt{2}$  (B)  $6\sqrt{2}$  (C)  $3\sqrt{3}$  (D)  $6\sqrt{3}$
17. In  $\triangle ABC$ ,  $\angle A = 90^\circ$  and I is the incentre. The perpendicular distance of I from BC is  $\sqrt{8}$ . Then AI is equal to  
 (A)  $\sqrt{8}$  (B) 3 (C)  $\sqrt{12}$  (D) 4
18. In an isosceles triangle, the centroid, the orthocentre, the incentre and the circumcentre are  
 (A) coincident (B) collinear  
 (C) in the interior of the circumcircle (D) in the interior of the incircle
19. If  $a, b$  are positive real numbers and  $\sqrt{a \frac{a}{b}} = a \cdot \sqrt{\frac{a}{b}}$  where  $a \frac{a}{b}$  is a mixed fraction, which of the following is true ?  
 (A)  $b = a^2 + 1$  (B)  $a = b^2 - 1$  (C)  $a = b^2 + 1$  (D)  $b = a^2 - 1$
20. If  $\frac{p}{a} + \frac{q}{b} + \frac{r}{c} = 1$  and  $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0$  then the value of  $\frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2}$  is  
 (A) 0 (B) -11 (C) 9 (D) 1
21. Let  $[x]$  denote the greatest integer less than or equal to  $x$ , what is the value of  $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{2004}]$ ?  
 (A) 58850 (B) 59730 (C) 59950 (D) 56718
22. If the roots of the equation  $x^2 - 2ax + a^2 + a - 3 = 0$  are real and less than 3 then  
 (A)  $a < 2$  (B)  $2 \leq a \leq 3$  (C)  $3 < a \leq 4$  (D)  $a > 4$
23. If a function  $f(x)$  is defined such that  $10^{f(x)} = \frac{1-x}{1+x}$  where  $x < 1$ , then  $f(a) + f(b)$  is equal to  
 (A)  $f\left(\frac{a+b}{1+ab}\right)$  (B)  $f\left(\frac{a-b}{1+ab}\right)$  (C)  $f\left(\frac{a-b}{1-ab}\right)$  (D) none of these
24. How many solutions are there for  $(a, b)$  if  $7ab73$  is a five digit number divisible by 99 ?  
 (A) 3 (B) 2 (C) 0 (D) 1
25. The number  $107^{90} - 76^{90}$  is divisible by  
 (A) 61 (B) 62 (C) 64 (D) none of these

26. A sequence  $a_0, a_1, a_2, a_3, \dots, a_n \dots$  is defined such that  $a_0 = a_1 = 1$  and  $a_{n+1} = (a_{n-1} \cdot a_n) + 1$  for  $n \geq 1$ . Which of the following is true ?
- (A)  $4 \mid a_{2004}$       (B)  $3 \mid a_{2003}$       (C)  $5 \mid a_{2004}$       (D)  $2 \mid a_{2003}$
27. A solid cuboid has edges of length  $a, b, c$ . What is the surface area ?
- (A)  $(a + b + c)^2 - (a^2 + b^2 + c^2)$       (B)  $abc$   
 (C)  $2(a^2 + b^2 + c^2)$       (D)  $ab + bc + ca$
28. A circle and a parabola are drawn on a piece of paper. The number of regions they divide the paper into is at most
- (A) 3      (B) 4      (C) 5      (D) 6
29. A cubic polynomial  $P$  is such that  $P(1) = 1, P(2) = 2, P(3) = 3$  and  $P(4) = 5$ . Then  $P(6)$  is
- (A) 7      (B) 10      (C) 13      (D) 16
30. Which of the following is the best approximation to  $\frac{(2^3 - 1)(3^3 - 1)(4^3 - 1) \dots (1000^3 - 1)}{(2^3 + 1)(3^3 + 1)(4^3 + 1) \dots (1000^3 + 1)}$  ?
- (A)  $\frac{3}{5}$       (B)  $\frac{33}{50}$       (C)  $\frac{333}{500}$       (D)  $\frac{3333}{5000}$

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